FIELD OBSERVATION REPORT

Common Effluent Treatment Plant (CETP)

Delhi, India

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ABSTRACT

This report presents a detailed scientific analysis of a Common Effluent Treatment Plant (CETP) located in Delhi, India, visited as part of an environmental chemistry field study. The plant, commissioned in 2002 and operated by a consortium of industrial companies under government supervision, treats a mixed influent of industrial wastewater and domestic sewage at a capacity of approximately 3,000 m³ per day. This report examines each unit process in rigorous chemical and physico-chemical detail — including coagulation–flocculation chemistry, disinfection mechanisms, adsorption theory, and sedimentation dynamics — and critically evaluates both the plant's performance against regulatory effluent standards and its significant operational limitations. Key findings include marked reductions in BOD (200 → 17.8 mg/L), COD (600 → 117.4 mg/L), and TSS (250–300 → 19.4 mg/L), although TDS remains high (~2,000 mg/L) and the treated effluent is unsuitable for potable or recreational use. Recommendations for process upgrading, including Moving Bed Biofilm Reactor (MBBR) installation and reverse osmosis, are discussed.

Keywords: *common effluent treatment plant; coagulation; flocculation; chlorination; activated carbon adsorption; sludge management; wastewater chemistry; BOD; COD; TDS.*

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1. Introduction and Contextual Background

A Common Effluent Treatment Plant (CETP) is a centralised wastewater management facility designed to collect, treat, and discharge effluent generated by multiple industrial and, in some cases, domestic sources. Unlike individual effluent treatment plants (ETPs) that serve a single industry, a CETP enables economies of scale, shared operational costs, and centralised monitoring — making it a critical instrument of industrial environmental compliance, particularly in developing urban–industrial contexts such as Delhi, India.

The plant visited during this field study was commissioned by the Delhi Government in 2002, forming part of a network of 13 CETPs distributed across the city. The majority of these facilities, including the one under observation, were constructed between 2000 and 2005 by Larsen & Toubro (L&T), one of India's foremost engineering and construction conglomerates. The plant is presently managed by a registered society of industrial companies under direct government supervision and employs approximately 15 operational personnel.

Although the plant was originally engineered and dimensioned exclusively for industrial effluent, its influent composition has changed substantially since commissioning. It now also receives domestic sewage from approximately 2,000 makeshift dwellings (jhuggi clusters) in adjacent areas. This admixture introduces significantly elevated concentrations of biodegradable organic compounds, nitrogenous species (ammonia, urea, nitrates), and faecal pathogens — substantially altering the influent chemistry and challenging the design assumptions of the existing treatment train.

> **Note:** *The influx of domestic sewage into an industrially-designed CETP represents a common urban environmental management failure in rapidly developing cities, where informal settlements proliferate faster than formal sewage infrastructure. This creates a mismatch between design intent and operational reality, with significant implications for treatment efficacy and regulatory compliance.*

1.1 Objectives of the Field Visit

- To observe and document each stage of the treatment process in operation.
- To understand the chemical and physico-chemical mechanisms underpinning each unit process.
- To critically evaluate the plant's performance by comparing influent and effluent quality parameters.
- To identify operational limitations and propose scientifically grounded improvements.

2. Plant Overview and Influent Characterisation

2.1 Operational Parameters

The following table summarises the key design and operational parameters of the CETP as observed during the field visit:

Parameter	Value / Description
Year Commissioned	2002
Constructed By	Larsen & Toubro (L&T)
Operating Body	Industrial Society (Government Supervised)

Parameter	Value / Description
Personnel	~15
Total Influent Volume	~3,000 m³/day
Outflow Rate	~185 m³/hour
Influent Sources	Industrial effluent + Domestic sewage (~2,000 households)

Table 2.1 — Summary of CETP operational parameters as observed during field visit.

2.2 Influent Water Quality Characterisation

The physicochemical composition of the raw influent is a product of its dual origin: industrial processes generating refractory organics, heavy metals, and high dissolved solids, and domestic sources contributing biodegradable organics, suspended faecal matter, and nitrogenous compounds. The following table presents the measured influent quality parameters:

Parameter	Measured Influent Value	Indian CPCB Discharge Standard	Significance
BOD (mg/L)	~200	≤30	High organic load from domestic sewage
COD (mg/L)	~600	≤250	Refractory organics from industrial sources
TSS (mg/L)	250–300	≤100	Suspended solids from both sources
TDS (mg/L)	2,500–3,000	≤2,100	High salt load from industrial processes
pH	~8	6.5–8.5	Mildly alkaline; within acceptable range
Heavy Metals	Elevated (varied)	Limits per metal	From electroplating/dyeing industries
Pathogens	Present	MPN < 1000/100 mL	From domestic sewage component

Table 2.2 — Influent water quality parameters with regulatory benchmarks (CPCB, India).

2.3 The BOD:COD Ratio as an Indicator of Treatability

A critical parameter in wastewater characterisation is the ratio of Biochemical Oxygen Demand (BOD) to Chemical Oxygen Demand (COD). BOD measures the oxygen consumed by microbial degradation of biodegradable organics over a 5-day incubation period at 20°C, while COD measures the total oxygen demand of all oxidisable matter (both biodegradable and refractory) via chemical oxidation with potassium dichromate.

$$BOD_5 : COD \text{ ratio} = 200 : 600 \approx 0.33$$

A BOD:COD ratio of approximately 0.33 for this influent indicates a moderately biodegradable wastewater stream. A ratio above 0.5 is generally considered easily biodegradable; values below 0.3 indicate a high proportion of refractory (chemically oxidisable but biologically resistant) compounds — common in industrial effluents from dyeing, tanning, and electroplating industries.

The value of 0.33 reflects the mixed industrial–domestic nature of the influent, where refractory industrial compounds dilute the more biodegradable domestic sewage fraction.

3. Preliminary Treatment Processes

Preliminary treatment removes gross pollutants — floating materials, large solids, and heavy settleable particles — that would otherwise impair downstream mechanical and chemical processes. These stages involve primarily physical separation mechanisms and do not yet alter the dissolved chemical composition of the effluent.

3.1 Flow Regulation Chamber

On entering the plant, the raw influent passes first through a flow regulation chamber. This unit serves as a hydraulic buffer: it attenuates peak flow rates arising from diurnal variation in wastewater generation (characteristically peaking in the early morning and early evening) to produce a more uniform, steady-state flow to downstream units.

Hydraulic stability is chemically critical because chemical dosing in downstream coagulation units is calibrated to a specific volumetric flow rate. Uncontrolled flow surges cause underdosing during peak flow — producing inadequate coagulation — and overdosing during low-flow periods — causing reagent wastage and potential pH disruption. The flow regulation chamber thus maximises the efficiency of chemical dosing by reducing coefficient of variation in flow.

3.2 Screening

3.2.1 Oil and Grease Screen

The first screening unit targets the removal of free-floating oil, grease, and petroleum hydrocarbons. These are removed by mechanical rods or a rotating drum skimmer that physically lifts the hydrophobic floating layer from the surface of the wastewater.

The chemical rationale for early oil removal is significant: oils and greases adsorb onto the surface of colloidal particles in the wastewater, forming a hydrophobic coating that interferes with the electrostatic destabilisation mechanisms of subsequent coagulation. Specifically, oil films can shield negatively charged colloidal surfaces from coagulant cations (Fe^{3+}), thereby reducing coagulation efficiency. Furthermore, oils can foul the filter media in the Dual Media Filter and reduce the adsorption capacity of the Activated Carbon Filters by blocking pore sites.

3.2.2 Bar Screen (20 mm Aperture)

A manually cleaned bar screen with 20 mm apertures intercepts coarse solid material — plastics, rags, vegetation, packaging material — entrained in the influent stream. This is a purely physical filtration step operating on the principle of size exclusion.

Though chemically trivial in isolation, effective bar screening is indispensable for downstream process protection: large fibrous materials can entangle pump impellers, block pipe elbows, and blind the surfaces of tube settlers and filter media. Inadequate gross solids removal exponentially increases maintenance frequency and downtime, disrupting the continuity of chemical treatment.

3.3 Grit Removal Unit

Following screening, the wastewater enters a grit chamber — a widened channel designed so that the forward velocity drops sufficiently to allow dense inorganic particles (sand, silt, gravel, grit; specific gravity ~2.65) to settle by gravity while keeping lighter organic solids in suspension. A mechanical scraper periodically sweeps accumulated grit to a sump, from which a screw conveyor transports it for landfill disposal.

The governing principle is Stokes' Law of sedimentation, which describes the terminal settling velocity (v_s) of a spherical particle in a viscous fluid:

$$v_s = [d^2(\rho_p - \rho_f) \times g] / (18\mu)$$

where d is particle diameter (m), ρ_p and ρ_f are the densities of particle and fluid respectively (kg/m³), g is gravitational acceleration (9.81 m/s²), and μ is dynamic viscosity of the fluid (Pa·s). Grit particles, being dense (ρ_p ≈ 2,650 kg/m³) and relatively coarse (d > 0.2 mm), settle rapidly, allowing their selective removal at flow velocities of approximately 0.3 m/s — insufficient to settle lighter, less dense organic flocs.

> **Note:** *Grit removal protects downstream pumps and pipes from abrasive wear, prevents accumulation of inert material in biological reactors (where applicable), and ensures that TSS measurements at subsequent sampling points reflect only organic and colloidal material rather than inorganic sand — critical for accurate process monitoring.*

4. Equalisation and Homogenisation

4.1 Equalisation Tank — Function and Chemical Importance

The equalisation tank is one of the most operationally critical units in the CETP. Raw industrial–domestic wastewater is characterised by extreme diurnal variability in both flow rate and pollutant concentrations. Industrial discharge events — such as a batch dump from a dyeing or electroplating operation — can momentarily spike COD, heavy metal concentrations, and pH far beyond their daily averages. If such a spike entered the dosing system unattenuated, it would overwhelm coagulant stoichiometry and disrupt the flocculation chemistry downstream.

The tank is equipped with continuous submerged aeration — diffused air is bubbled through the tank contents — which serves three simultaneous functions:

- Prevention of quiescent settling: ensures that settleable organic solids and colloidal particles remain in suspension, preventing premature sedimentation and septic zone formation within the tank.
- Mixing and homogenisation: mechanically blends influents of varying composition and concentration into a time-averaged, compositionally uniform stream for consistent downstream chemical dosing.
- Prevention of anaerobic conditions: maintains dissolved oxygen (DO) levels above the critical threshold (~0.5 mg/L), preventing the growth of sulphate-reducing bacteria (SRB) that generate hydrogen sulphide (H_2S) — a corrosive, malodorous, and toxic gas — and methanogenic archaea.

$$SO_4^{2-} + 8H^+ + 8e^- \rightarrow H_2S + 4H_2O \quad \text{(anaerobic sulphate reduction)}$$

The equalisation tank thus acts as a chemical stabilisation buffer — reducing variance in pH, BOD, COD, and TSS to a manageable range — so that the dosing calculations performed by operators remain valid across all operational conditions.

5. Chemical Dosing and Primary Treatment

Primary chemical treatment at this CETP involves the sequential addition of three chemical agents to destabilise colloidal particles, form settleable flocs, and inactivate pathogenic micro-organisms. Each chemical performs a specific and distinct electrochemical or biochemical function.

5.1 The Colloidal Stability Problem

Most pollutants in both industrial and domestic wastewater exist as colloidal particles — particles in the size range of approximately 1 nm to 1 µm — that are too small to settle under gravity in any practical timeframe but too large to be removed by dissolved-phase treatment. Colloidal particles in wastewater are predominantly negatively charged, arising from the dissociation of surface functional groups (carboxylic, phenolic, sulphate groups) and selective adsorption of anions.

This negative surface charge creates a repulsive electrostatic barrier — quantified by the zeta potential (ζ) — that prevents particles from approaching sufficiently close for Van der Waals attractive forces to cause aggregation. The DLVO theory (Derjaguin, Landau, Verwey, and Overbeek) describes the balance between these repulsive electrostatic and attractive Van der Waals forces as a function of inter-particle distance. To achieve coagulation, the repulsive energy barrier must be overcome by reducing the zeta potential toward zero — achieved through the addition of coagulants.

> **Note:** *Zeta potential (ζ) is a measure of the electrostatic potential at the shear plane surrounding a colloidal particle. Stable colloids typically have ζ values greater than ±30 mV. Coagulation is optimised when ζ is reduced to within ±5 mV, indicating charge neutralisation.*

5.2 Ferric Alum — Coagulation Chemistry

Ferric alum (iron(III) sulphate, $Fe_2(SO_4)_3$) is the primary coagulant, dosed at 400–500 mg/L — a significantly high dose reflecting the elevated turbidity and colloidal load of the mixed influent. Upon dissolution in water, ferric ions undergo a complex series of hydrolysis reactions:

$$Fe^{3+} + 3H_2O \rightarrow Fe(OH)_3\downarrow + 3H^+$$

$$Fe^{3+} + OH^- \rightarrow FeOH^{2+} \rightarrow Fe(OH)_2^+ \rightarrow Fe(OH)_3\downarrow$$

The ferric hydroxide precipitate ($Fe(OH)_3$) is an amorphous, gelatinous solid with a very large surface area. It operates by two simultaneous mechanisms:

- Charge neutralisation: Fe^{3+} and its partially hydrolysed cationic species ($FeOH^{2+}$, $Fe(OH)_2^+$) adsorb onto negatively charged colloidal particle surfaces, reducing the zeta potential and overcoming the repulsive electrostatic barrier.
- Sweep flocculation (enmeshment): the precipitating $Fe(OH)_3$ 'sweeps' colloidal particles from suspension as it settles, physically enmeshing them within the growing floc matrix — an especially effective mechanism at the relatively high alum doses used here.

The hydrolysis reactions are pH-sensitive: $Fe(OH)_3$ solubility is minimised (and coagulation efficiency maximised) in the pH range of approximately 6.5–8.0. At the observed influent pH of ~8, coagulation conditions are near-optimal for ferric species.

The consumption of OH^- in hydrolysis (and production of H^+ in the first equation above) has the secondary effect of slightly reducing the pH of the treated water, though the plant's alkaline influent provides sufficient buffering capacity to maintain pH within acceptable limits.

$$\text{Alkalinity buffering: } HCO_3^- + H^+ \rightarrow H_2CO_3 \rightarrow H_2O + CO_2\uparrow$$

5.3 Poly-Electrolytes — Flocculation Chemistry

Poly-electrolytes (also termed polyelectrolytes or polymer flocculants) are high-molecular-weight synthetic polymers dosed at approximately 1 mg/L (ppm) following the primary coagulant. They are available in anionic, cationic, and non-ionic forms; the specific type used at this plant was not disclosed, though anionic or non-ionic poly-electrolytes are most common for secondary wastewater flocculation.

Their mechanism of action is bridging flocculation: the extended polymer chains adsorb simultaneously onto two or more micro-floc particles, physically linking them into a larger, denser, more rapidly settling macro-floc. This is distinct from the charge-neutralisation mechanism of

ferric alum and is complementary to it: the alum produces primary micro-flocs by charge destabilisation, and the polymer then bridges these micro-flocs into settleable aggregates.

> **Note:** *The dose of poly-electrolyte must be carefully controlled: underdosing produces insufficient bridging and inadequate floc formation; overdosing can cause restabilisation — where excess polymer molecules adsorb fully onto individual particles, restoring steric repulsion and preventing aggregation. Jar testing is the standard laboratory method used to determine the optimal coagulant and polymer doses.*

5.4 Bleaching Powder — Disinfection Chemistry

Bleaching powder (calcium hypochlorite, $Ca(OCl)_2$), also commonly written as $Ca(ClO)_2$, is dosed at approximately 20 mg/L as a primary disinfectant. On dissolution in water, it releases hypochlorite ion:

$$Ca(OCl)_2 + 2H_2O \rightarrow Ca^{2+} + 2OCl^- + 2H_2O$$

$$OCl^- + H^+ \rightleftharpoons HOCl \quad (pK_a \approx 7.54 \text{ at } 25^\circ C)$$

At the observed pH of approximately 8, the Henderson-Hasselbalch relationship predicts that hypochlorite ion (OCl^-) predominates over hypochlorous acid (HOCl). This is pharmacologically important: HOCl is the biologically active species and is approximately 80–100 times more effective as a bactericide than OCl^-. Its neutral charge allows it to penetrate bacterial cell membranes and oxidatively damage essential enzymes, nucleic acids, and structural proteins through a mechanism of electrophilic chlorination.

At pH 8, approximately 72% of total chlorine exists as OCl^- and only ~28% as HOCl. This reduced biocidal efficacy at mildly alkaline pH is partially compensated by the extended contact time provided in the Pre-Chlorination Contact Tank (PCCT).

Beyond microbial inactivation, chlorine dosing at this stage provides an additional benefit: oxidation of soluble organic compounds. HOCl is a powerful oxidant (standard reduction potential $E^\circ = +1.49$ V) that can mineralise certain organics or convert them to more oxidised, polar forms that are more amenable to subsequent coagulation and adsorption.

> **Note:** *One significant concern with chlorination of wastewater containing natural organic matter (NOM) is the formation of disinfection by-products (DBPs), including trihalomethanes (THMs, e.g., chloroform $CHCl_3$) and haloacetic acids (HAAs). These are carcinogenic compounds formed when HOCl reacts with humic and fulvic acids. Since the treated effluent from this CETP is not intended for potable use, DBP formation is a lower regulatory priority than at drinking water treatment plants, but it remains an environmental concern for the receiving watercourse.*

6. Mixing, Flocculation, and Sedimentation

6.1 Pre-Chlorination Contact Tank (PCCT)

The pre-chlorination contact tank receives additional bleaching powder dosing and provides a regulated hydraulic retention time (HRT) during which the chlorine-wastewater contact is extended. The CT value (concentration × time, units mg·min/L) governs the degree of pathogen inactivation according to Chick–Watson kinetics:

$$\ln(N/N_0) = -k \times C^n \times t$$

where N and N_0 are the final and initial microbial concentrations respectively, k is a rate constant specific to the micro-organism and disinfectant, C is the disinfectant concentration (mg/L), n is the dilution coefficient, and t is contact time (min). Increasing t in the PCCT directly improves log reduction of pathogens.

6.2 Flash Mixer

The flash mixer is a tank equipped with high-speed mechanical impellers operating at relatively high rotational velocities (typically generating velocity gradients G of 300–1,000 s^{-1}). Its sole function is rapid, turbulent mixing of the coagulant solution throughout the entire volume of wastewater to ensure stoichiometrically uniform contact between Fe^{3+} ions and colloidal particles. The mixing time is deliberately brief (typically 30–60 seconds) to prevent excessive shear that would disrupt nascent micro-floc formation.

The velocity gradient G (Camp number) quantifies the intensity of mixing:

$$G = \sqrt{(P / \mu V)}$$

where P is the power input to the mixer (W), μ is dynamic viscosity (Pa·s), and V is tank volume (m^3). Rapid mixing (high G) is used in the flash mixer; slow mixing (low G, 20–75 s^{-1}) is used in the subsequent flocculation tank to allow gentle aggregation without floc shearing.

6.3 Flocculation Tank

Following rapid mixing, the wastewater passes to the flocculation tank, where gentle, low-shear paddle mixing promotes the collision and aggregation of the destabilised micro-flocs into larger, more settleable macro-flocs. The poly-electrolyte added upstream plays a critical role here, providing the bridging linkages that hold the growing floc aggregate together against the gentle shear forces. The Camp number G is deliberately reduced to 20–75 s^{-1}, and the hydraulic retention time is extended (typically 15–30 minutes) to allow sufficient inter-particle collisions for floc growth.

The rate of particle aggregation is governed by Smoluchowski's perikinetic and orthokinetic collision models. Orthokinetic (fluid-shear-driven) collisions dominate at the floc sizes achieved in this tank. The collision rate is proportional to the product of the velocity gradient G, the volume fraction of particles φ, and the diameter of the colliding particles. This explains why the optimal flocculation velocity gradient decreases as floc size increases: at low G, large, low-density flocs can aggregate without shear disruption; at high G, floc strength is exceeded and the aggregate fractures.

6.4 Tube Settler

The tube settler is an enhanced sedimentation unit that exploits the principle of shallow-depth sedimentation to achieve a vastly higher effective settling area within a compact footprint. It consists of banks of inclined polypropylene or PVC tubes (typically 50–100 mm diameter, inclined at 45–60° to the horizontal), through which the flocculated wastewater flows upward in a laminar regime.

The inclined surfaces reduce the effective settling depth that a floc particle must traverse before reaching a solid surface. The Reynolds number within the tubes (Re = ρvD/μ) is maintained well below 500 (laminar flow), ensuring that particle trajectories are governed by gravity rather than turbulent eddies. The settling efficiency η is described by:

$$\eta = 1 - \exp(-v_s \times L\cos(\theta) / [Q/W \times (L\sin(\theta) + D)])$$

where v_s is Stokes settling velocity, L is tube length, θ is inclination angle, Q/W is the surface overflow rate, and D is the tube diameter. In practice, tube settlers can achieve surface overflow rates 5–10 times higher than conventional sedimentation tanks for equivalent removal efficiencies.

Accumulated sludge slides down the inclined tube surfaces under gravity and collects in a hopper at the base, while the clarified supernatant overflows from the top of the settler into the filtration train. The sludge underflow is pumped to the sludge treatment system.

7. Filtration Processes

7.1 Dual Media Filter (DMF)

The dual media filter (DMF) is the first of two filtration stages and represents a well-established engineered improvement over conventional single-medium (sand) filters. The plant operates nine DMF tanks in parallel, providing redundancy and allowing individual tanks to be taken offline for backwashing without interrupting overall treatment.

7.1.1 Media Configuration and Principle of Operation

Each tank contains three stratified layers of filter media, arranged in order of decreasing particle size and decreasing density from top to bottom:

Layer	Medium	Particle Size (mm)	Density (g/cm³)	Primary Function
Top	Anthracite	1.0–2.0	~1.5	Captures large floc; low headloss
Middle	Quartz Sand	0.4–0.8	~2.65	Fine particle capture; structural stability
Bottom	Gravel	2.0–5.0	~2.65	Support layer; flow distribution

Table 7.1 — DMF media stratification, properties, and function.

The lower density of anthracite (1.5 g/cm³) relative to sand (2.65 g/cm³) is critical: during backwashing, hydraulic fluidisation causes the media to stratify by density, naturally returning the coarser anthracite to the top and fine sand to the middle. This self-restoring stratification allows the filter to be backwashed and returned to service without manual re-stratification.

Filtration occurs by three distinct mechanisms operating simultaneously:

- Straining: particles larger than the inter-grain pore spaces are physically excluded. This is the dominant mechanism for larger suspended flocs in the anthracite layer.
- Adsorption: smaller particles adsorb onto grain surfaces via van der Waals forces and electrostatic attraction. The residual positive surface charge on iron floc particles promotes electrostatic adsorption onto the quartz sand (negatively charged silica surface).
- Biological activity: in filters operating continuously, a biological film (schmutzdecke in sand filters; less developed in DMFs) can develop on grain surfaces, providing some biodegradation of dissolved organics.

As filtration proceeds, head loss across the media increases as captured particles fill inter-grain void spaces. When head loss reaches the design maximum (typically 1.5–2.0 m), backwashing is initiated. At this plant, backwashing is performed approximately once per week. During backwashing, a combination of air scouring (from a blower) and an upward water flow (from a backwash pump) is used to fluidise and expand the media bed, dislodging trapped particles. The backwash water, heavily laden with suspended solids, is returned to the equalisation tank for re-treatment — a standard closed-loop process water recovery practice.

> **Note:** *The DMF removes TSS only. It has no meaningful capacity to remove dissolved pollutants such as COD, ammonia, or TDS. The transition from effluent TSS of 250–300 mg/L (influent) to a post-tube-settler intermediate of perhaps 30–50 mg/L, and ultimately to a post-DMF value approaching the observed final effluent TSS of 19.4 mg/L, reflects the cumulative effectiveness of the coagulation–sedimentation–filtration sequence.*

7.2 Activated Carbon Filter (ACF)

The activated carbon filter (ACF) bank consists of six tanks operating in parallel, each packed with granular activated carbon (GAC). Activated carbon is produced by the controlled carbonisation and activation (steam or CO_2 treatment at 800–1,000°C) of carbonaceous precursors (typically coal, coconut shell, or wood), producing a highly porous material with internal surface areas of 500–1,500 m²/g — the defining property that makes it an extraordinary adsorbent.

7.2.1 Adsorption Mechanisms

Adsorption onto activated carbon is primarily a physical process (physisorption), driven by van der Waals and London dispersion forces between the sorbate molecules and the carbon surface, which is graphitic in structure and hydrophobic in character. The hydrophobic surface selectively attracts organic molecules over polar water molecules — a phenomenon of particular value for removing colour bodies, odour compounds, and residual dissolved organics that have resisted prior treatment.

The adsorption capacity of activated carbon is described by the Freundlich isotherm (empirically determined for most sorbate–sorbent systems):

$$q_e = K_F \times C_e^{(1/n)}$$

where q_e is the mass of sorbate adsorbed per unit mass of adsorbent at equilibrium (mg/g), C_e is the equilibrium concentration of sorbate in solution (mg/L), K_F is the Freundlich adsorption capacity parameter, and 1/n is the Freundlich intensity parameter (typically 0.1–0.5 for favourable adsorption). Alternatively, the Langmuir isotherm applies when adsorption occurs as a monolayer on a finite number of homogeneous surface sites.

The ACF removes:

- Colour: from residual azo dyes, humic acids, and iron-complexed organics from dyeing and textile industries.
- Odour: adsorbs volatile organic compounds (VOCs) including mercaptans, phenolics, and chlorinated organics.
- Residual organics: dissolved organics that passed through coagulation and DMF, reducing effluent COD further.
- Residual chlorine: activated carbon catalytically decomposes hypochlorous acid, which is important for controlling disinfection by-product formation.

$$C + 2HOCl \rightarrow CO_2 + 2HCl \quad \text{(approximate catalytic dechlorination)}$$

Over time, the adsorption capacity of the GAC bed becomes exhausted as all available surface sites are occupied. The media must then either be regenerated (thermally reactivated at high temperature) or replaced. The frequency of regeneration/replacement was not specified during the visit but is typically governed by effluent colour and COD breakthrough.

7.3 Post-Chlorination Contact Tank (Post-PCCT)

A final chlorination stage is applied after filtration to ensure residual disinfection of the fully filtered effluent before discharge. At this point, the treated water is substantially free of TSS and residual organics that would exert chlorine demand, meaning that the dosed chlorine is available primarily for microbial inactivation rather than oxidation of organic matter. A measurable residual chlorine concentration (typically 0.5–1.0 mg/L free residual chlorine) is maintained to provide ongoing disinfection protection in the discharge drainage system.

The bottom layer of the post-PCCT (more concentrated in retained particulates) is recycled back to the beginning of the process. The top layer constitutes the final treated effluent. Continuous online monitoring instruments at this point transmit real-time quality data (pH, COD, turbidity) to government environmental authorities — a mandatory regulatory compliance measure for CETPs in India.

8. Treated Effluent Quality — Analysis and Discussion

The following table presents a direct comparison of influent and treated effluent quality parameters, alongside regulatory discharge standards:

Parameter	Influent (mg/L)	Effluent (mg/L)	Removal Efficiency	CPCB Standard	Compliance?
BOD	~200	17.8	91.1%	≤30	✓ Yes
COD	~600	117.4	80.4%	≤250	✓ Yes
TSS	250–300	19.4	~93%	≤100	✓ Yes
TDS	2,500–3,000	~2,000	~20%	≤2,100	✗ Marginal
pH	~8	~8	—	6.5–8.5	✓ Yes

Table 8.1 — Influent vs. effluent water quality comparison with regulatory compliance assessment.

8.1 BOD Removal Analysis

BOD removal of 91.1% — from approximately 200 to 17.8 mg/L — is achieved within the regulatory discharge limit of 30 mg/L and reflects the effectiveness of the coagulation–flocculation–sedimentation sequence in removing colloidal and suspended organic matter. However, it is important to note that the physical–chemical process train at this CETP achieves BOD reduction primarily through the removal of particulate and colloidal BOD (associated with suspended organic particles captured by coagulation) rather than through biodegradation of dissolved BOD. The residual effluent BOD of 17.8 mg/L is attributable largely to soluble, non-coagulable dissolved organic matter that has passed through the entire treatment train.

8.2 COD Removal Analysis

COD removal of 80.4% — from ~600 to 117.4 mg/L — represents satisfactory but not exceptional performance. The residual effluent COD of 117.4 mg/L, while below the CPCB limit of 250 mg/L, indicates a substantial load of refractory dissolved organics that were not removed by coagulation, filtration, or carbon adsorption. This is consistent with the complex industrial effluent background, which may contain recalcitrant compounds such as azo dye chromophores, phenolic compounds, and chlorinated hydrocarbons that resist physical–chemical treatment.

The COD:BOD ratio of the effluent is approximately 117.4 / 17.8 ≈ 6.6 — compared to 3.0 in the influent. This dramatic increase in the COD:BOD ratio confirms that the treatment process has preferentially removed biodegradable organic matter (reflected in BOD), leaving behind a proportionally much higher fraction of refractory, non-biodegradable organics (reflected only in COD). This has implications for the receiving watercourse: though the absolute COD is reduced, the remaining organic load is largely resistant to natural biodegradation and may persist in the environment.

8.3 TDS — The Critical Limitation

TDS reduction from approximately 2,500–3,000 mg/L to ~2,000 mg/L represents only a ~20% removal — the most significant performance gap of the plant. This is chemically expected: TDS comprises dissolved ionic species (Na^+, K^+, Ca^{2+}, Mg^{2+}, Cl^-, SO_4^{2-}, NO_3^-, HCO_3^-, and various

heavy metal ions) and dissolved organics that pass through all physical and chemical barriers employed at this plant, including coagulation, sedimentation, filtration, and activated carbon.

The residual TDS of ~2,000 mg/L approaches but may marginally exceed the CPCB discharge standard of 2,100 mg/L. More critically, the WHO guideline for drinking water TDS is 500 mg/L (desirable) with an upper tolerance of 1,000–1,500 mg/L. The effluent TDS is thus 4–5× above potable standards. Effective TDS reduction requires membrane-based treatment technologies, specifically Reverse Osmosis (RO):

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Osmotic pressure: π = iMRT  (van't Hoff equation for dilute
                            solutions)
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In RO, a hydraulic pressure exceeding the osmotic pressure of the feed solution (typically 10–30 bar for brackish water) is applied across a semi-permeable membrane, forcing water molecules through while rejecting dissolved ions. RO can reduce TDS from 2,000 mg/L to below 100 mg/L — well within potable standards. However, the permeate (product water) is virtually devoid of essential minerals and would require remineralisation (addition of Ca^{2+}, Mg^{2+}, HCO_3^-) before human consumption. The capital and operational cost of RO, and management of the resulting concentrated brine reject, represent significant barriers to its implementation at this plant.

8.4 Assessment: Is the Effluent Safe?

Based on the quality parameters observed, the treated effluent from this CETP is emphatically not potable and cannot be considered safe for bathing, primary contact recreation, or agricultural irrigation without further treatment. The effluent is discharged into a communal stormwater drain, where it mixes with untreated wastewater from surrounding informal settlements and unregulated industrial discharges — a situation that severely undermines the pollution control benefit achieved by the CETP.

9. Sludge Generation, Treatment, and Disposal

9.1 Sludge Composition and Hazard Classification

Sludge is generated at multiple points in the process train — as underflow from the tube settler, as backwash residuals from the DMF and ACF, and as screenings from the bar screen and grit chamber. The chemical composition of the sludge is heterogeneous and reflects the complexity of the mixed influent stream:

- Heavy metals (Cr, Pb, Cd, Ni, Cu, Zn) concentrated by coagulation from electroplating and galvanising effluents
- Precipitated iron hydroxide from ferric alum dosing
- Adsorbed petroleum hydrocarbons and organic dyes
- Residual polymeric coagulant flocculant
- Inorganic grit and sand
- High TDS from concentrated dissolved salts

This composition means that the sludge meets the criteria for classification as a hazardous waste under India's Hazardous Waste (Management, Handling and Transboundary Movement) Rules, 2008 — a classification that imposes strict legal requirements for its handling, transport, and disposal.

9.2 Sludge Dewatering

Raw sludge from the tube settler underflow has a very high water content (typically 95–99.5% by mass), making direct disposal economically and logistically prohibitive. Dewatering is

achieved at this plant using cloth membrane filtration under hydraulic compression (a plate-and-frame or belt filter press type arrangement), operated as a batch process. This reduces the water content to approximately 60–80%, producing a dewatered sludge cake of substantially reduced volume that is more amenable to transport and disposal.

9.3 TCLP Analysis — Toxicity Assessment

Before disposal, representative samples of the dewatered sludge cake are subjected to the Toxicity Characteristic Leaching Procedure (TCLP) — a standardised leachate test (US EPA Method 1311, adopted in modified form by Indian regulatory practice) designed to simulate the leaching behaviour of waste materials in a landfill environment.

In the TCLP procedure, the sludge sample is contacted with a dilute acetic acid extraction fluid (pH ≈ 4.93, simulating acidic landfill leachate conditions) at a liquid:solid ratio of 20:1, agitated for 18 hours, and filtered through a 0.6–0.8 µm glass fibre membrane. The filtrate (the 'TCLP leachate') is then analysed for approximately 40 parameters including:

- Heavy metals (Ag, As, Ba, Cd, Cr, Hg, Pb, Se) by Atomic Absorption Spectroscopy (AAS)
- Volatile and semi-volatile organic compounds (VOCs, SVOCs) by GC-MS
- Chlorinated pesticides and herbicides
- Specific inorganic anions

The TCLP results determine the legal disposal pathway for the sludge cake:

Sludge Treatment Stage	Description
Sludge Collection	Collected from tube settler underflow, DMF backwash, and ACF cleaning cycles
Thickening	Gravity concentration; free water drained. Sludge cake forms.
Cloth Membrane Filtration	Further dewatering using cloth press; hydraulic compression applied in batch cycles
TCLP Analysis	~40 parameters tested for leachability of heavy metals and toxic organics
Disposal Pathway	Hazardous landfill (₹1,000/ton) | Brick kiln addition | Controlled combustion

Table 9.1 — Sludge treatment stages and disposal pathway options.

Disposal at a permitted hazardous waste landfill costs approximately ₹1,000 per tonne. Co-processing in cement kilns or brick manufacturing is a preferred option where TCLP results permit, as it reduces landfill burden and recovers calorific value. Improper disposal — a significant risk given enforcement challenges — creates long-term groundwater contamination risk through leachate migration.

10. Summary Process Flow Diagram

The following diagram presents the complete treatment train in sequential order, showing the stage number, unit name, and key operational details at each step:

1	INFLUENT INTAKE	Industrial effluent + domestic sewage ~3,000 m³/day | pH ~8 | BOD ~200 mg/L
	↓	
2	FLOW REGULATION CHAMBER	Reduces hydraulic turbulence. Ensures uniform flow to downstream units.
	↓	
3	SCREENING & OIL REMOVAL	20 mm bar screen (manual) removes gross solids. Mechanical oil/grease skimmer removes hydrocarbons.
	↓	
4	GRIT REMOVAL UNIT	Gravity settling of heavy inorganic particles (sand, silt). Screw conveyor transports grit for disposal.
	↓	
5	EQUALISATION TANK	Continuous aeration prevents settling & anaerobic conditions. Homogenises flow and pH variations.
	↓	
6	CHEMICAL DOSING	Poly-electrolyte (~1 ppm) | Bleaching powder (~20 ppm) Ferric alum (~400–500 ppm)
	↓	
7	PRE-CHLORINATION CONTACT TANK (PCCT)	Extended contact time for disinfection. Oxidation of soluble organics by HOCl.
	↓	
8	FLASH MIXER	High-speed mechanical agitation. Ensures uniform coagulant dispersion.
	↓	
9	FLOCCULATION TANK	Gentle paddle mixing promotes floc growth. Poly-electrolytes bridge micro-flocs into settleable masses.
	↓	
10	TUBE SETTLER	Inclined-tube laminar flow enhances settling. Sludge blanket forms; supernatant decanted from top.
	↓	
11	DUAL MEDIA FILTER (DMF)	9 tanks | Layers: Anthracite → Sand → Gravel. Removes residual TSS. Backwash weekly.
	↓	
12	ACTIVATED CARBON FILTER (ACF)	6 tanks | Physical adsorption. Removes colour, odour, residual organics.
	↓	
13	POST-CHLORINATION CONTACT TANK	Final disinfection. Online monitoring. Treated effluent discharged to communal drain.

Figure 10.1 — CETP sequential treatment process flow, from raw influent to final effluent discharge.

11. Operational Problems and Critical Limitations

11.1 Absence of Biological Treatment

The most scientifically significant deficiency of this CETP is the complete absence of biological treatment processes. The current process train is entirely physical–chemical in character. Biological treatment — the controlled utilisation of micro-organisms to metabolise dissolved organic compounds — is essential for the removal of dissolved BOD, ammonia, and nitrogenous compounds that are not removed by coagulation or filtration.

In particular, the nitrogenous load introduced by the domestic sewage component (in the form of ammoniacal nitrogen, NH_4^+) is not addressed. Ammonia exerts a significant nitrogenous oxygen demand (NOD) in receiving watercourses, where it is oxidised by nitrifying bacteria in a two-stage process:

$$NH_4^+ + 1.5O_2 \rightarrow NO_2^- + H_2O + 2H^+ \quad \text{(Nitrosomonas sp.)}$$

$$NO_2^- + 0.5O_2 \rightarrow NO_3^- \quad \text{(Nitrobacter sp.)}$$

Each gram of ammoniacal nitrogen requires 4.57 g of dissolved oxygen for complete nitrification — a significant additional oxygen demand that depletes DO in the receiving watercourse and damages aquatic ecosystems. This is not currently controlled by the plant.

11.2 TDS — Absence of Membrane Treatment

As discussed in Section 8.3, TDS cannot be reduced by physical–chemical processes and requires membrane technology (RO). The absence of RO means that the plant is fundamentally unable to produce effluent meeting potable water TDS standards, and that high concentrations of dissolved salts — including potentially toxic heavy metal ions — are discharged continuously.

11.3 Infrastructure Degradation

Several infrastructure issues were observed or reported during the site visit:

- Electrical control panels: outdated and potentially non-compliant with current safety standards; risk of electrical faults affecting dosing pump operation.
- Pump failures: frequent mechanical failures noted; critical for continuous chemical dosing and flow management.
- Pipe corrosion: extensive use of mild steel (MS) pipes in a chemically aggressive environment (alternating acid/alkali, chlorinated water, abrasive grit) leads to accelerated corrosion, leakage, and potential contamination of treated water. Cast iron (CI), ductile iron (DI), or stainless steel (SS) pipes would provide substantially greater service life.

11.4 Mixing of Industrial and Domestic Streams

The admixture of domestic sewage into a system designed solely for industrial effluent creates a fundamental design mismatch. The domestic fraction introduces a variable, high-BOD, high-nitrogen load that the physical–chemical process train cannot efficiently treat, while simultaneously diluting the coagulable industrial particulate load. A dedicated biological stage (see Section 12) is essential to address the domestic sewage component.

12. Recommendations for Process Improvement

12.1 Moving Bed Biofilm Reactor (MBBR)

The most impactful process upgrade would be the installation of a Moving Bed Biofilm Reactor (MBBR) between the equalisation tank and the chemical dosing stage. In an MBBR, freely suspended polyethylene carrier elements (specific gravity ~0.95) are kept in continuous motion within an aerated reactor by the combined action of the aeration air and the fluid turbulence.

Biofilm — dense microbial communities dominated by heterotrophic bacteria for BOD removal and autotrophic nitrifiers for ammonium oxidation — grows attached to the protected internal surfaces of the carriers.

The MBBR offers several advantages over conventional activated sludge:

- No secondary clarifier required: biofilm-coated carriers remain in the reactor; only sloughed biomass requires downstream settling.
- Compact footprint: high biofilm surface area per unit reactor volume (200–500 m^2/m^3 depending on carrier fill fraction).
- Robustness to hydraulic and organic shock loads: the biofilm community is less vulnerable than suspended sludge to transient toxic industrial discharges.
- Simultaneous nitrification: with appropriate carrier media and aerobic conditions, ammoniacal nitrogen can be oxidised to nitrate — addressing the NOD problem described in Section 11.1.

12.2 Reverse Osmosis for TDS Control

Installation of a partial-flow RO system on the final treated effluent stream would reduce TDS to below 100 mg/L in the permeate stream, which could be blended with bypass effluent to produce a final effluent TDS in the range of 300–500 mg/L — within potable standards. The concentrate (brine reject) stream would require careful management, including potential evaporation, crystallisation, or permitted disposal.

12.3 Infrastructure Modernisation

- Replace all MS pipework with schedule 80 PVC or ductile iron (DI) pipes rated for CETP chemical service.
- Install modern programmable logic controller (PLC)-based dosing systems with online flow, pH, and turbidity feedback to enable automatic dosing rate adjustment.
- Replace aging electrical panels with modern MCC (Motor Control Centre) units compliant with IS standards.
- Implement SCADA (Supervisory Control and Data Acquisition) monitoring for real-time process visibility and remote alarm notification.

13. Analytical Methods for Effluent Quality Monitoring

The following analytical tests may be conducted on influent and effluent samples to characterise and monitor treatment performance. These tests are described with their underlying chemical principles:

Parameter	Analytical Method	Chemical Principle	Typical Detection Range
BOD_5	5-Day Incubation (IS 3025 Part 44)	Winkler titration for DO depletion by aerobic microbial oxidation at 20°C	0–300 mg/L
COD	Dichromate Reflux (IS 3025 Part 58)	$Cr_2O_7^{2-}$ oxidises organics in hot H_2SO_4; excess Cr^{6+} titrated with FAS	10–800 mg/L
TSS	Gravimetric (IS 3025 Part 17)	Sample filtered through 1.2 µm GF/C; residue dried at 105°C and weighed	5–500 mg/L

Parameter	Analytical Method	Chemical Principle	Typical Detection Range
TDS	Gravimetric (IS 3025 Part 16)	Filtrate evaporated at 180°C; dissolved residue weighed	50–5,000 mg/L
pH	Potentiometric Electrode (IS 3025 Part 11)	Nernst equation: $E = E° + (RT/nF)\ln[H^+]$; glass membrane electrode	0–14
Residual Chlorine	DPD Colorimetry (IS 3025 Part 26)	N,N-diethyl-p-phenylenediamine reacts with HOCl → pink colour; spectrophotometric	0.02–5 mg/L
Heavy Metals	Atomic Absorption Spectroscopy (AAS)	Ground-state atoms absorb light at characteristic wavelengths; Beer-Lambert law	µg/L–mg/L
Ammoniacal N	Indophenol Blue (IS 3025 Part 34)	NH_3 + phenol + hypochlorite → indophenol blue complex (640 nm)	0.05–1.0 mg/L
Nitrate-N	UV Spectrophotometry / Cadmium Reduction	NO_3^- reduced to NO_2^- by Cd; diazotization with sulfanilamide (540 nm)	0.01–10 mg/L
Phosphate	Molybdenum Blue (IS 3025 Part 31)	PO_4^{3-} + molybdate + ascorbic acid → phosphomolybdenum blue (880 nm)	0.01–5 mg/L
TCLP Metals	AAS / ICP-OES on TCLP leachate	As for heavy metals; after standard leaching extraction procedure	µg/L–mg/L

Table 13.1 — Analytical methods for monitoring CETP influent and effluent quality.

14. Conclusion

This field observation and subsequent analysis demonstrates that the Delhi CETP under study achieves substantial reductions in conventional wastewater quality parameters — particularly BOD (91.1% removal), COD (80.4%), and TSS (~93%) — through a well-designed sequence of physical–chemical processes. The coagulation chemistry of ferric alum, the electrostatic and bridging mechanisms of poly-electrolytes, the kinetics of hypochlorous acid disinfection, the fluid dynamics of tube settlers, and the adsorptive capacity of granular activated carbon collectively constitute a robust, well-engineered treatment system that brings the effluent within regulatory compliance for the majority of key parameters.

The plant illustrates a well-integrated application of environmental chemistry principles across its treatment train, from the hydraulic buffering of the flow regulation chamber through to the final polishing achieved by activated carbon adsorption. Each unit process performs a distinct, complementary role, and the redundancy built into the filtration banks (nine DMF tanks and six ACF tanks operating in parallel) reflects a design that supports continuous, reliable operation. The staged chlorination strategy, applied both before and after filtration, provides layered disinfection assurance for the treated stream.

The CETP also plays an important institutional role: it centralises treatment for a network of industrial dischargers under government supervision, enabling shared infrastructure, economies of scale, and coordinated environmental compliance monitoring that would be far harder to achieve through individual facility-level treatment. The plant's continuous online monitoring of

pH, COD, and turbidity, transmitted to government environmental authorities, further exemplifies its role in transparent regulatory oversight.

From an educational and scientific standpoint, this field visit offered valuable first-hand insight into the practical application of coagulation–flocculation theory, disinfection kinetics, adsorption isotherms, and sedimentation dynamics within an operating industrial-scale facility. Observing these unit processes in sequence provided a clear demonstration of how fundamental physico-chemical principles translate into effective, real-world wastewater management, underscoring the plant's continuing value both as a pollution-control asset and as a site of applied environmental chemistry.

End of Report